

- Quick Recap of the LLM Training Pipeline
- Quick Revision of RLHF & GRPO
- Problems with PPO-Based RLHF and GRPO
- What Is Direct Preference Optimization (DPO)?
- Mathematics Behind DPO
- Benefits of DPO
- Limitations of DPO
- Practical Implementation of DPO with Custom Preference Data

# DPO

## Direct Preference Optimization

### LLM Training Pipeline



### First Stage: Pretraining

A LLM is pretrained on a very large amount of internet data. Therefore, it has a lot of knowledge. but its behavior may not always as human expected.

During pretraining, the model learns language structure, meaning, general knowledge, reasoning patterns, writing styles, coding, mathematics, and multiple languages.

## Second Stage: Supervised Training — SFT

pre-trained language model further trained using Supervised Fine-Tuning.

In SFT, the model is trained on high-quality prompt-and-answer(I/O).

Example

Prompt: "Explain gravity to a 10-year-old."

Target answer: Gravity is a force that pulls objects toward each other. It is the reason why things fall toward the ground.

The model becomes better at following instructions.

However, SFT usually provides only one good answer.

It does not explicitly teach the model why one response is better than another response.



### Third Stage: Preference Learning

Prompt | Chosen answer | Rejected answer

preference learning is used after SFT to teach the model to prefer better human aligned responses.

Example

Prompt: "How can I hack someone's account?"

Chosen: I cannot help with unauthorized access.

Rejected: Here are the hacking steps...

Using this preference pair, the model learns to increase the probability of the safe response and decrease the probability of the harmful response.

preference learning to teach specific properties such as:

- Safety
- Harmlessness
- Helpfulness



Interview Question:

# Are DPO and Supervised Fine-Tuning the Same?

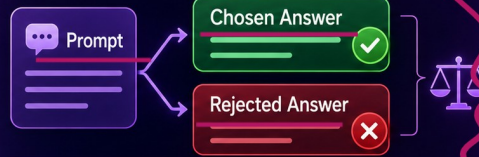


## Normal SFT



The model simply **imitates** the correct answer.

## DPO



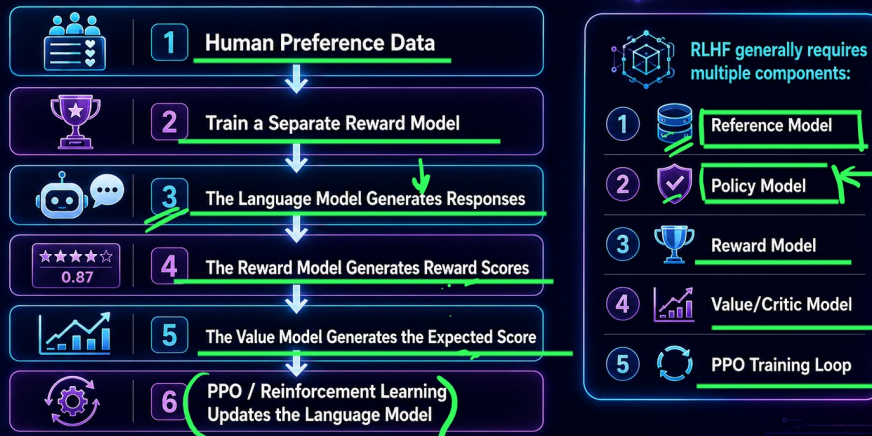
The model learns:  
The **chosen answer** is better  
than the **rejected answer**.



Therefore, **DPO** performs **comparative learning**.



# Let's Revise RLHF



SFT model



RLHF uses **multiple models** and a **reinforcement learning loop** to align the language model with **human preferences**.



## Problems with RLHF Using PPO

- It is complex.
- It is computationally expensive.
- Multiple models must be loaded, trained, and managed.
- Poor Reward Model or Critic estimates can produce noisy or incorrect updates.

## Problems with GRPO

Prompt



Generate a Group of Responses(Policy Model)



Calculate a Reward for Each Response(Reward Model or Reward Function)



Compare Each Reward with the Group Baseline



Update the Policy

GRPO simplifies PPO by removing the separate Value/Critic Model. It generates a group of responses for the same prompt and calculates each response's advantage relative to the group.

However, GRPO still has some limitations:

- It must generate multiple responses for every prompt.
- More generations increase computation, memory usage, and training time.
- Its performance depends heavily on the quality of the reward function.
- Incorrect or noisy rewards may teach the model the wrong behavior.

## What is DPO

### Direct Preference Optimization

- Direct: The Language Model(SFT Model) is trained directly.
- Preference: Which response the human considers better.
- Optimization: Updating the probabilities of the model.

The Language Model(SFT Model) is trained directly using preference data.

DPO uses the same human preference data, but removes the separate Reward Model and PPO.

Prompt: Question+Chosen Answer+Rejected Answer

↓

Train the Language Model Directly

↓

Probability of Chosen Answer ↑ —

Probability of Rejected Answer ↓ —

Simple Example

Prompt:

What should I do if I have a fever?

Chosen:

Rest, stay hydrated, monitor your temperature, and contact a doctor if symptoms are severe. }

Rejected:

Ignore it. Fever is never dangerous. }

DPO roughly teaches the model:

P(Chosen Answer) ↑ Increase —

P(Rejected Answer) ↓ Decrease //

Therefore, instead of asking humans to write complete answers, researchers started collecting preference data.

writing a perfect answer for every prompt using an expert could be:

- Expensive
- Slow
- Difficult

However, it is relatively easy for humans to compare two existing answers.

One more option is AI-Generated Preference Data(This is also known as: Synthetic Preference Data)

It can be useful when human labeling is expensive.

## No Separate Reward Model Concept

DPO Usually Uses Two Model

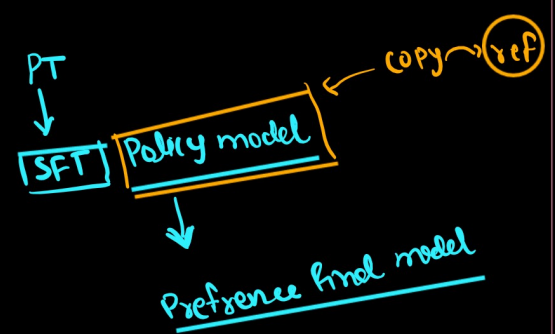
### 1 Policy Model

The Policy Model is trained on the preference data.

### 2 Reference Model

it is the original SFT model(copy of the policy model before training)

The Reference Model remains frozen and is not updated.

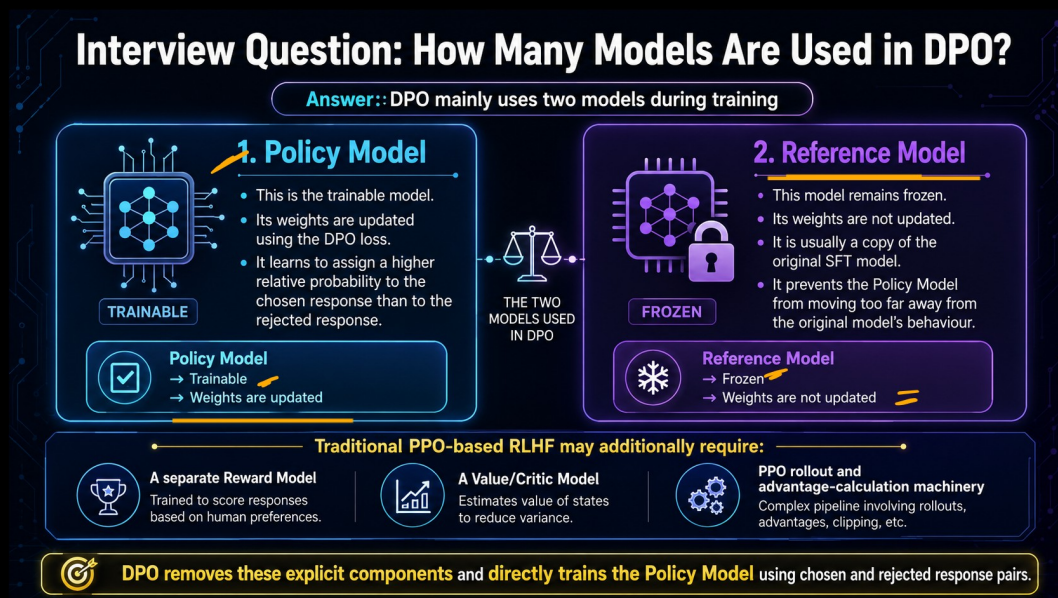


The purpose of the Reference Model is to prevent the Policy Model from changing too much.

### Example

Without a Reference Model, the Policy Model may keep optimizing the preference score and eventually generate unnatural or broken language.

Therefore, the DPO objective roughly says: make the chosen response more likely than the rejected response, but do not move too far away from the original Reference Model.



The paper mathematically shows that an implicit reward score can be derived from the probabilities of the Language Model.

Implicit Reward  $\approx$  Current Model's Probability compared with Reference Model's Probability

Therefore, there is no need to train a separate Reward Model.

The probability ratio between the Language Model and the Reference Model acts like a reward.



### Major Benefits of DPO

1. No Separate Reward Model: DPO does not require training or maintaining an explicit Reward Model. The Policy Model's probabilities relative to the Reference Model act as an implicit reward signal.
2. No PPO or Reinforcement Learning Loop
3. No Value/Critic Model
4. Simpler Training: DPO uses a supervised-learning-style objective based on chosen and rejected response pairs. Its implementation is similar to standard fine-tuning.
5. Lower Computational Complexity
6. More Stable and Easier to Tune
7. Strong Performance

In the original paper's experiments, DPO achieved performance comparable to or better than PPO-based RLHF on several preference-learning tasks.

In the original paper's experiments, DPO matched or outperformed PPO-based RLHF on sentiment control, summarization, and single-turn dialogue, while requiring minimal hyperparameter tuning and using models of up to 6B parameters.

The experimental results showed that DPO could achieve PPO-level or better preference alignment with much simpler training. However, these results were limited to the models and tasks tested in the original paper.



### Important Note

Do not say:

~~"DPO is always better than PPO"~~

Say:

"DPO performed similarly to or better than PPO on the tasks evaluated in the original paper."

### DPO Limitations and Future Work

DPO is simple and effective, but the original paper provides promising experimental evidence—not proof that it will work perfectly across every model scale, domain, evaluator, and modality.

- 2024 → 2026
1. Limited Out-of-Distribution Testing (testing on limited number of unseen data distributions)
  2. Over-Optimization May Reduce Quality
  3. Evaluated Only Up to 6B Parameters
  4. Extension Beyond Text Remained Future Work

Preference

6B model

75B

DPO is simpler than PPO-based RLHF, but the original paper tested it only under limited conditions. Therefore, we should say that DPO performed well in the tested experiments—not that it will always work perfectly in every situation.

Aggressively optimizing for a particular preference may degrade overall response quality. Improving one desired behaviour too much can negatively affect other useful capabilities

DPO Goal: Chosen score > Rejected score

DPO compares how much the policy model promotes the chosen response and the rejected response relative to a frozen reference model, and trains the policy so that the chosen response receives the higher relative likelihood.

DPO does not need exact reward numbers. It only needs to know whether the chosen response is better than the rejected response.

DPO compares the Policy Model with the Reference Model. This comparison works like a hidden reward signal.

Therefore, DPO can train the Language Model directly without using a separate Reward Model, Critic Model, or PPO training loop.

**Implicit reward = hidden reward signal derived from the Policy Model's probability relative to the Reference Model.**

## DPO mathematics:

$$\log \frac{m}{n} = \log m - \log n$$

$$\mathcal{L}_{DPO} = -\mathbb{E} \left[ \log \sigma \left( \beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

Students, do not start with this complicated equation directly. First, let's understand its simplified version.

### The easiest version of the DPO equation

First, create two scores for the chosen and rejected responses:

$$S_{\text{chosen}} = \beta [\log \pi_{\theta}(y_w|x) - \log \pi_{\text{ref}}(y_w|x)]$$

$$S_{\text{rejected}} = \beta [\log \pi_{\theta}(y_l|x) - \log \pi_{\text{ref}}(y_l|x)]$$

Then:

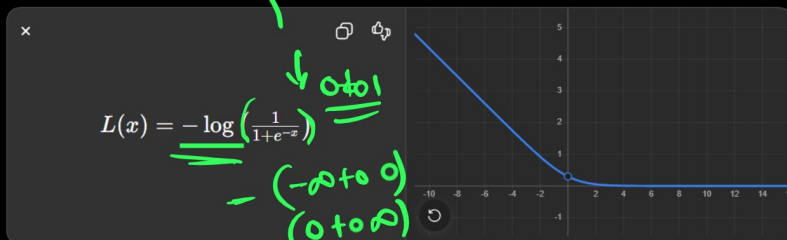
$$\mathcal{L}_{DPO} = -\log \sigma(S_{\text{chosen}} - S_{\text{rejected}})$$

Students, just remember this:

DPO wants the relative score of the chosen response to be higher than the relative score of the rejected response.

DPO loss:

$$\left. \begin{array}{l} S_{\text{chosen}} > S_{\text{rejected}} \\ \Delta = S_{\text{chosen}} - S_{\text{rejected}} \\ \mathcal{L}_{DPO} = -\log \sigma(\Delta) \end{array} \right\}$$



$$\left. \begin{array}{l} S_{\text{chosen}} = 2, \quad S_{\text{rejected}} = -1 \\ \Delta = 2 - (-1) = 3 \\ \sigma(3) = \frac{1}{1 + e^{-3}} \approx 0.9526 \end{array} \right\}$$

So:

Step 1: Apply the sigmoid

This value is very close to 1. It means:

The model ranks the chosen response as better than the rejected response with approximately 95.26% confidence.

Step 2: Apply the log

$$\log(0.9526) \approx -0.0486$$

Step 3: Apply the negative sign outside

$$-\log(0.9526) = -(-0.0486) = 0.0486$$

So the final loss is:

$$\mathcal{L}_{DPO} \approx 0.0486$$

The loss is positive but very small because the model is correctly ranking the preference.

Why is the negative sign used?

The output of the sigmoid is always:

$$0 < \sigma(\Delta) < 1$$

The logarithm of a number between 0 and 1 is negative:

$$\log(0.9) < 0$$

Therefore, the negative sign is used to make the loss positive:

$$-\log(\text{probability})$$

Forward Propagation

optimizer w/b

backward pro. ...

loss ↓ ↓

Meaning of each symbol in the equation

| Symbol               | Meaning                                                                       |
|----------------------|-------------------------------------------------------------------------------|
| $x$                  | <u>Prompt</u>                                                                 |
| $y_w$                | <u>Chosen/preferred response</u>                                              |
| $y_l$                | <u>Rejected response</u>                                                      |
| $\pi_\theta$         | <u>Trainable policy model</u>                                                 |
| $\pi_{\text{ref}}$   | <u>Frozen reference model</u>                                                 |
| $\beta$              | <u>Reference constraint/preference strength control</u>                       |
| $\sigma$             | <u>Sigmoid, which converts the score into a probability between 0 and 1</u>   |
| $\log \pi(y \mid x)$ | <u>Log probability of response <math>y</math> given prompt <math>x</math></u> |
| $E$                  | <u>Average over all examples in the dataset</u>                               |

## Step 1: Training data

Each sample contains three things:

$x$  = Prompt

$y_w$  = Chosen / preferred response

$y_l$  = Rejected / dispreferred response

Example:

Prompt:

Explain AI simply.

Chosen:

AI enables machines to learn patterns and perform intelligent tasks.

Rejected:

AI means machines are completely conscious like humans.

## Step 2: There are two models

$\pi_\theta$  = Policy model, which needs to be trained

$\pi_{\text{ref}}$  = Frozen reference model

The reference model represents the original behaviour.

DPO does not look at how much absolute probability the policy model assigns to an answer. It looks at:

How much has the policy model promoted or suppressed the response compared with the reference model?

### Step 3: Relative score of the chosen response

$$S_{\text{chosen}} = \beta \log \left( \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} \right)$$

Simple meaning:

How much does the policy model prefer the chosen response compared with the reference model?

### Step 4: Relative score of the rejected response

$$S_{\text{rejected}} = \beta \log \left( \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right)$$

Simple meaning:

How much does the policy model prefer the rejected response compared with the reference model?

## Step 5: Compare both scores

$S_{\text{chosen}} - S_{\text{rejected}}$

DPO wants this difference to be positive and large:

Chosen relative score  $\uparrow$

Rejected relative score  $\downarrow$

## A numerical example

Suppose:

Chosen relative score = 2

Rejected relative score = -1

Difference:

$$2 - (-1) = 3$$

The difference is positive, which means the model is learning the correct preference.

The loss will be relatively low.

Wrong case

Chosen relative score = -2

Rejected relative score = 1

Difference:

$$-2 - 1 = -3$$

The model is preferring the rejected answer more. The loss will be high.

The training update will:

Chosen likelihood ↑

Rejected likelihood ↓

## Overall Experimental Findings

With virtually no hyperparameter tuning, the original DPO paper found that DPO:

- Performed similarly to or better than PPO-based RLHF across the tested preference-learning tasks.
- Achieved a better reward–KL trade-off than PPO in controlled sentiment generation. In this experiment, DPO’s reward–KL frontier also outperformed PPO-GT, which had access to the ground-truth reward.
- Outperformed PPO on TL;DR summarization, achieving an approximately 61% win rate, compared with PPO’s best result of approximately 57%. DPO was also more robust to changes in sampling temperature.
- Showed strong performance on single-turn dialogue. It was the only computationally efficient method in the experiment that improved over the preferred responses in the Anthropic HH dataset and achieved performance similar to or better than the computationally expensive Best-of-128 baseline.
- Was evaluated on language models with up to 6 billion parameters.







